



Islamic University Of Technology

EEE 4308 : Digital Electronics

Project Name : SAP – 1

Team : Digital Electronauts

Phase - B

Team Advisor: Mr. Hayder Jahan Parash

Team Members:

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Objective:

- 1) Learning about Registers of SAP-1
- 2) Learning about Memory unit of SAP-1
- 3) Gaining knowledge about D-latch
- 4) Gaining knowledge about D Flip-flop
- 5) Working principle of RAM

Description:

This project is about understanding of 6 bit computer and construction of it. Basically its about understanding inside process of a computer. We can understand what does it mean clock speed of a computer and how we can control this speed. For running this computer we need to program the RAM first, then we can run the computer. Program is stored inside RAM as a set of instruction. Other modules acts according to instruction. Program counter counts the instruction and MAR sends the memory address for next instruction. Instructed operation is occurred inside ALU. Value of ALU is stored inside Accumulator. When necessary , Accumulator sends the data to output register to show the result. Necessary signal for all the module is sent by controller sequencer. Controller sequencer generates signals according to instruction code sent by instruction register. All the module is runed by clock pulse. Without clock pulse this computer can't work. There is HLT operation in our computer that stopes the main clock, as a result whole computer stopes. This is total working system of our SAP-1.

Phase - B

Registers:

1)Accumulator:

It is a 6-bit register which loads the data from RAM and sent it to ALU for necessary operation. It also stores the output value of ALU and sent in to Output register for showing the result.

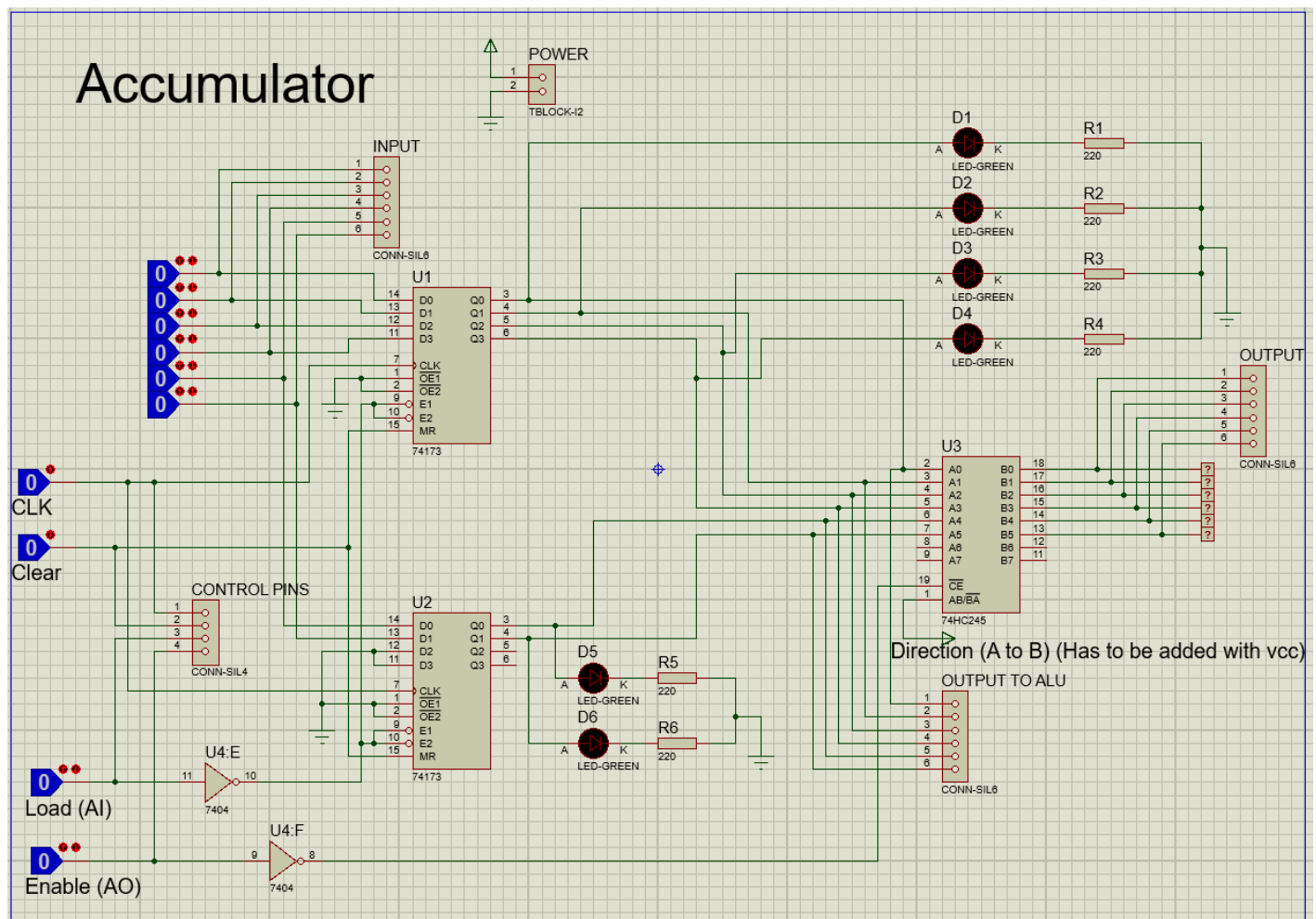
Control Bits:

- 1) **AI:** Stores value from BUS
- 2) **AO:** Sends its stored value to BUS

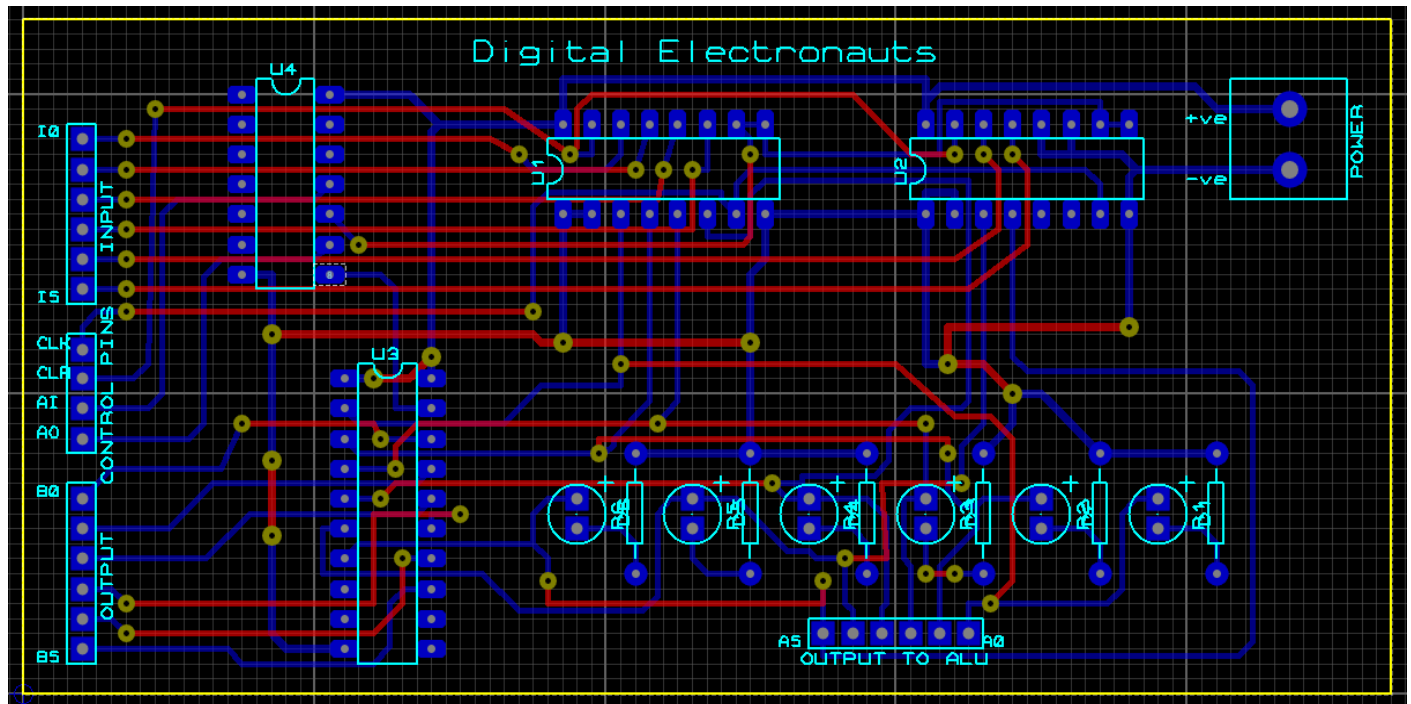
Circuit:

To built Accumulator we are using two 74LS173 IC's , one 74LS245 and one 7404 (NOT Gate). 74173 is a 4-bit register and 74245 is Octal BUS transreceiver (Tri-state Buffer).

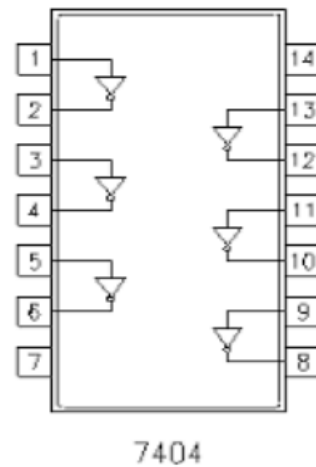
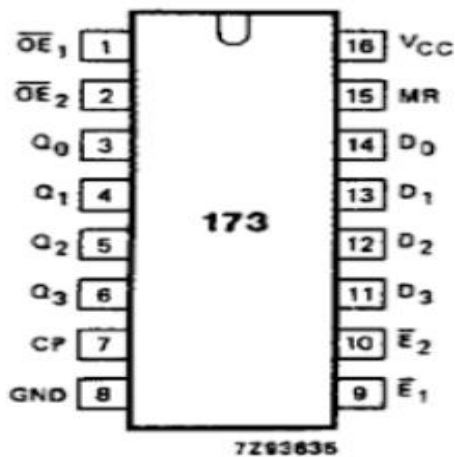
Accumulator Circuit in Proteous:



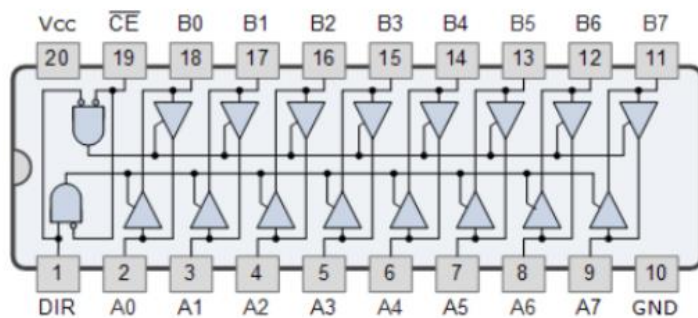
PCB :



Used IC's :



74LS245 Octal Tri-state Buffer



2) B-Register:

It is also a 6-bit register which loads the data from RAM and sent it to ALU for necessary operation.

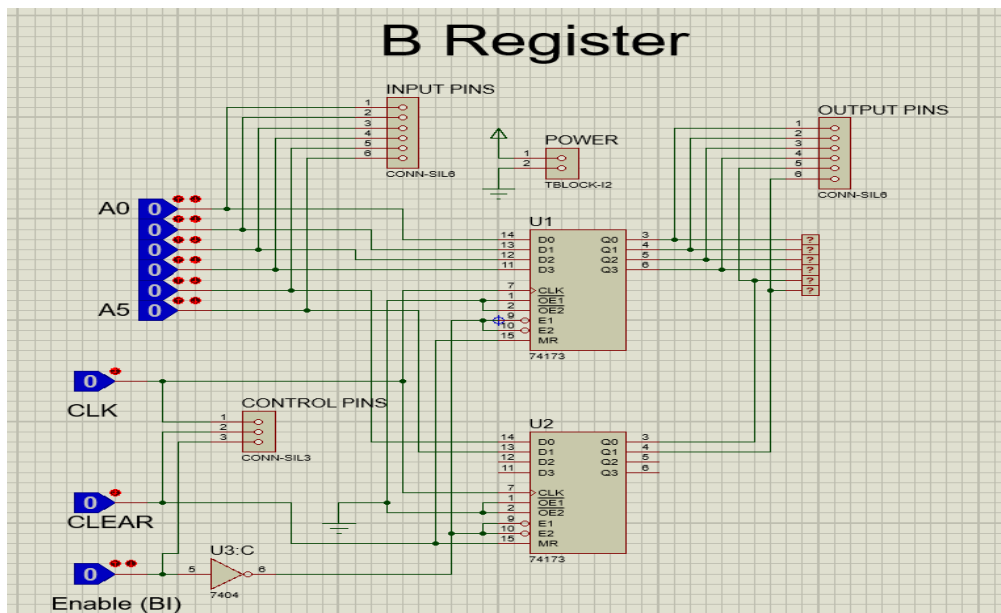
Control Bits:

- 1) **BI:** Stores Data from BUS

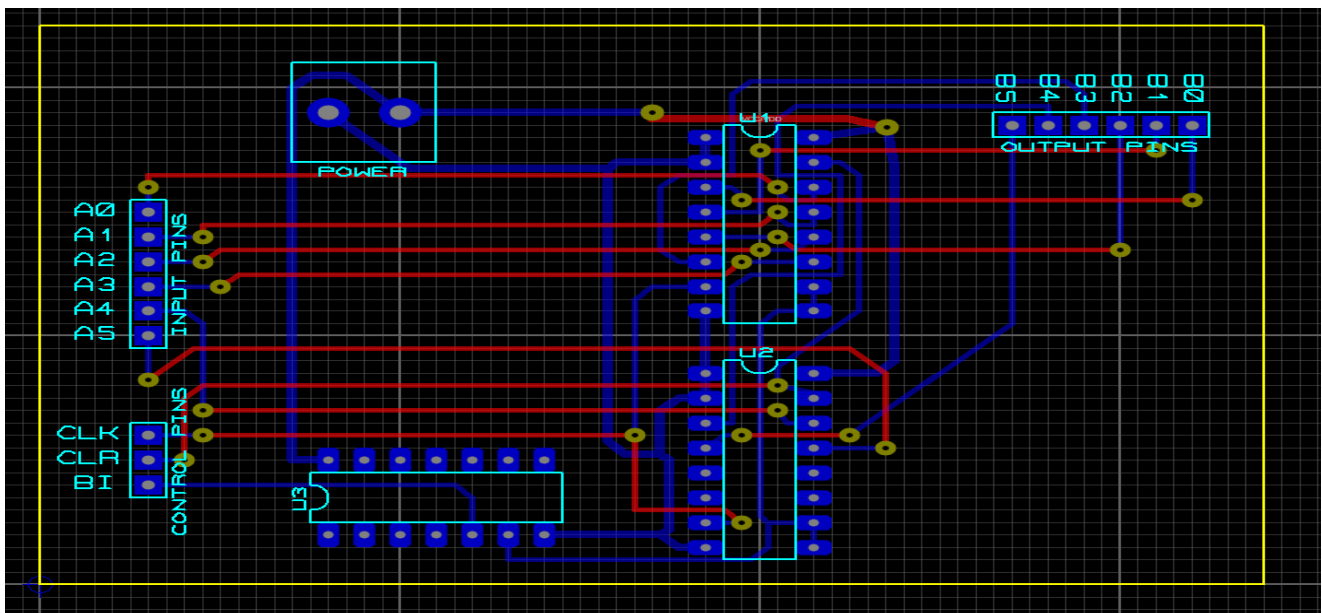
Circuit:

To built B-register we are using two 74LS173 IC's and one 7404 (NOT Gate). 74173 is a 4-bit register.

Circuit in Proteous:



PCB:



3) Output Register:

It is also a 6-bit register which loads the data from BUS and sent it to Display unit to show the results.

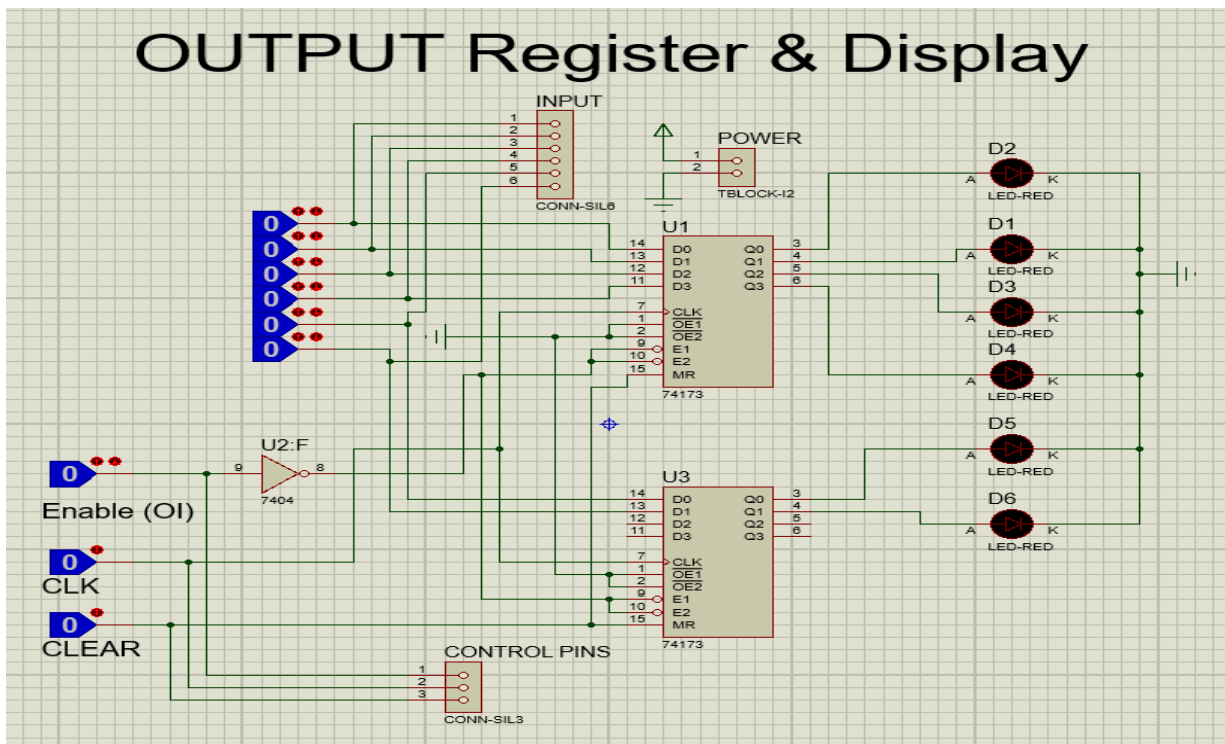
Control Bits:

- 1) **OI:** Sends stored value to display unit

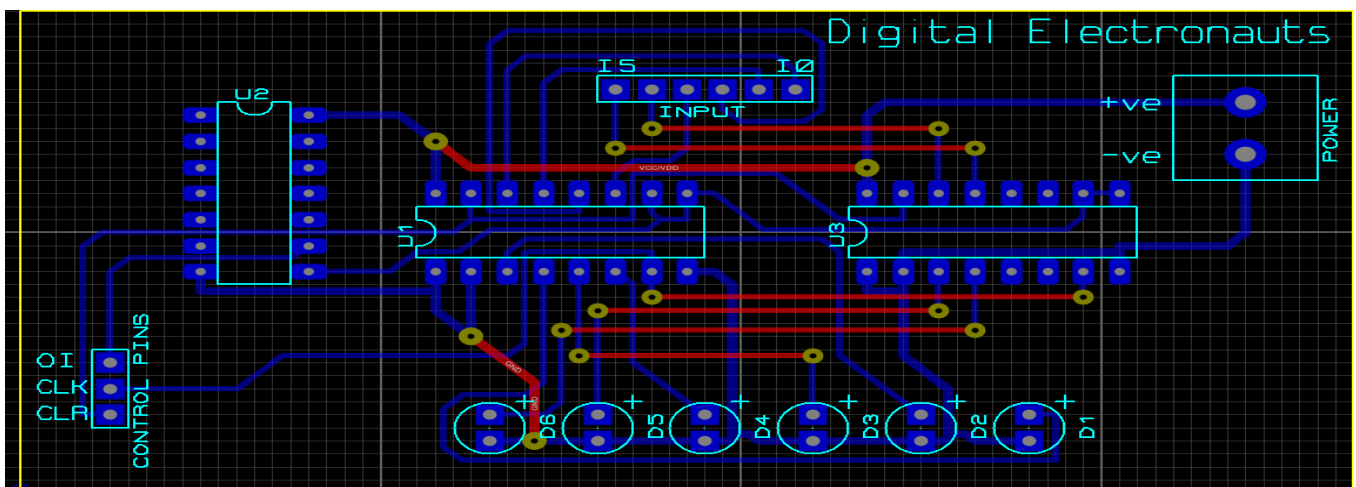
Circuit:

To built B-register we are using two 74LS173 IC's and one 7404 (NOT Gate). 74173 is a 4-bit register.

Circuit in Proteous:



PCB:



Memory Unit:

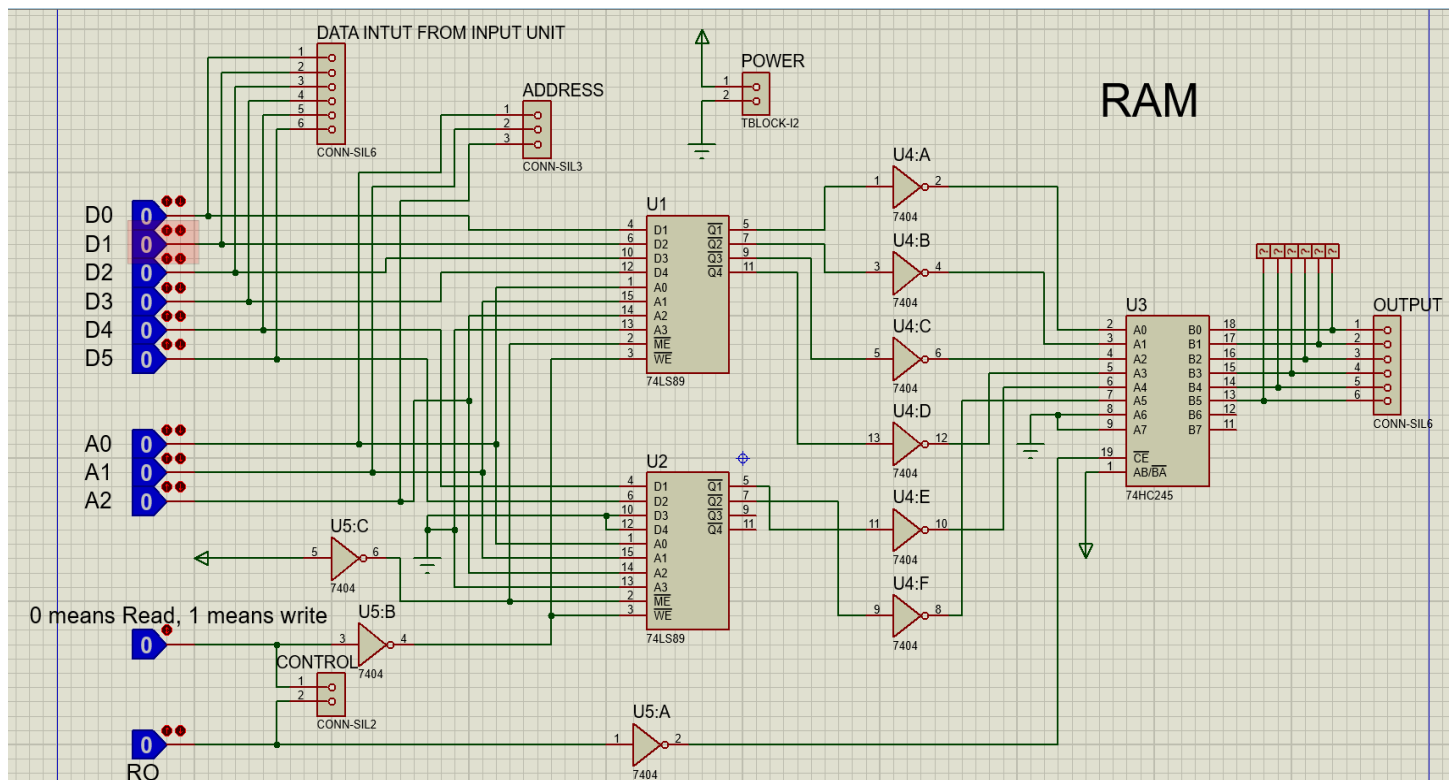
RAM:

It is one of the important module of the SAP-1. It is the program unit, here instructions are stored. Before running the SAP we need to program the RAM, taking it to the PROG store. Our RAM is 8×6 bit, it means there is 8 memory location and each location can store 6 bits of Data. When in run mode it takes memory address from MAR. Then it sends the stored value in that specific location to BUS.

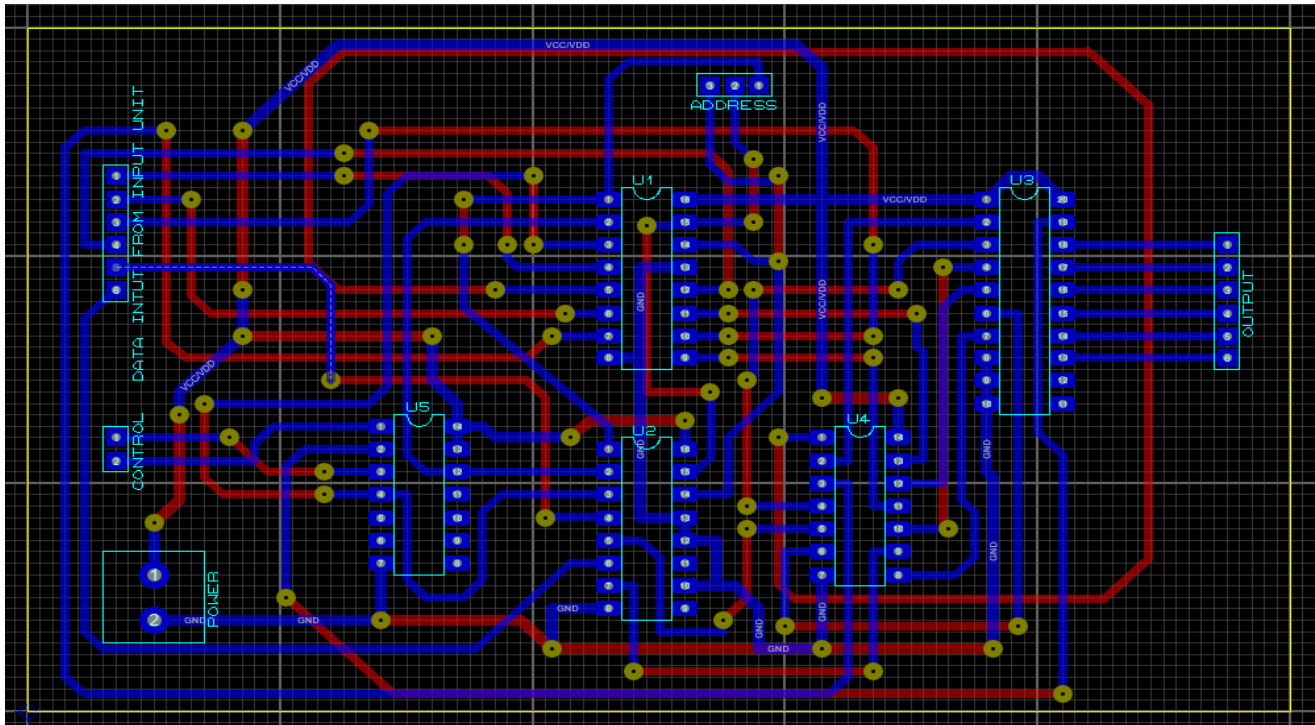
Control Bits:

- 1) **RO**: Puts current value to BUS
- 2) **CS**: Select pin
- 3) **R/W** : Read or write mode
- 4) **ADRS** : Accessing RAM location

Circuit in Proteous:

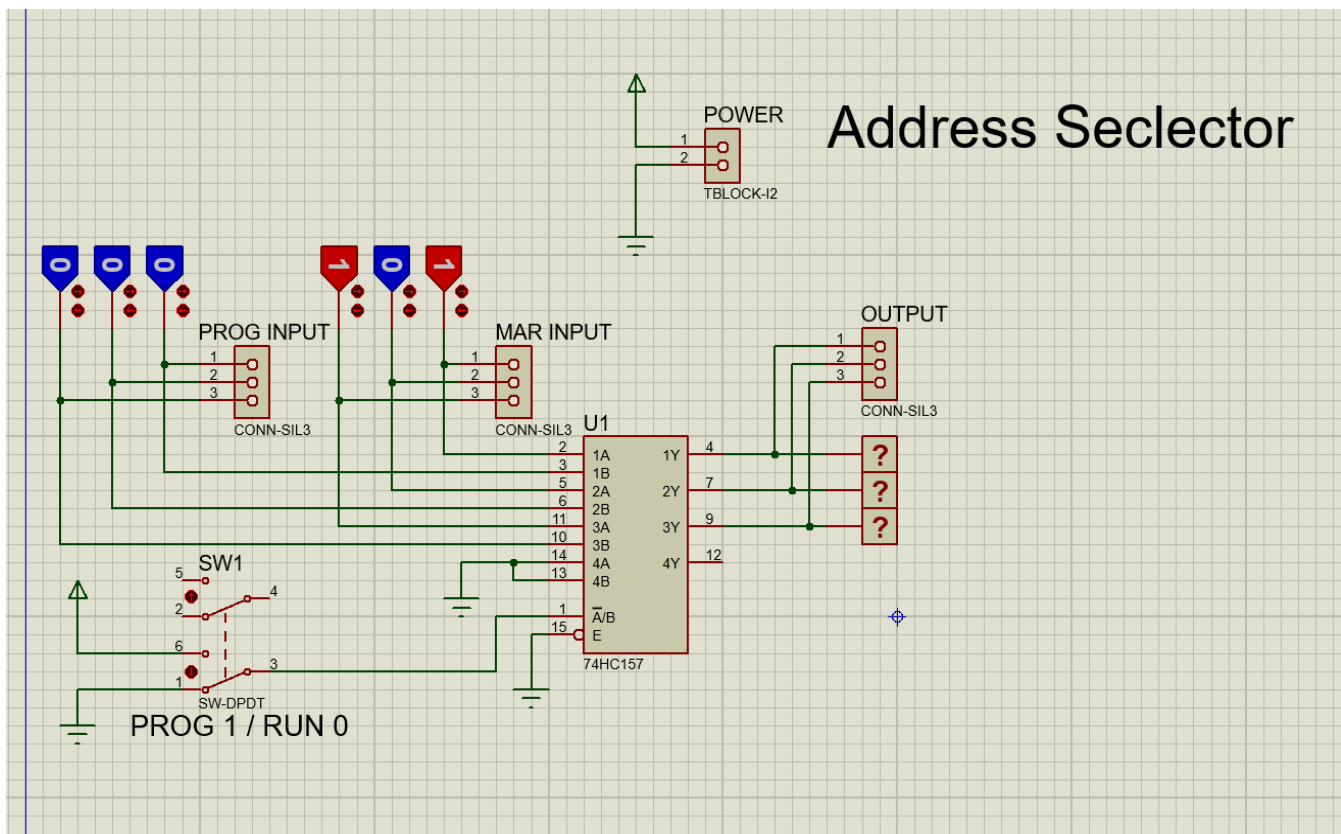


PCB:

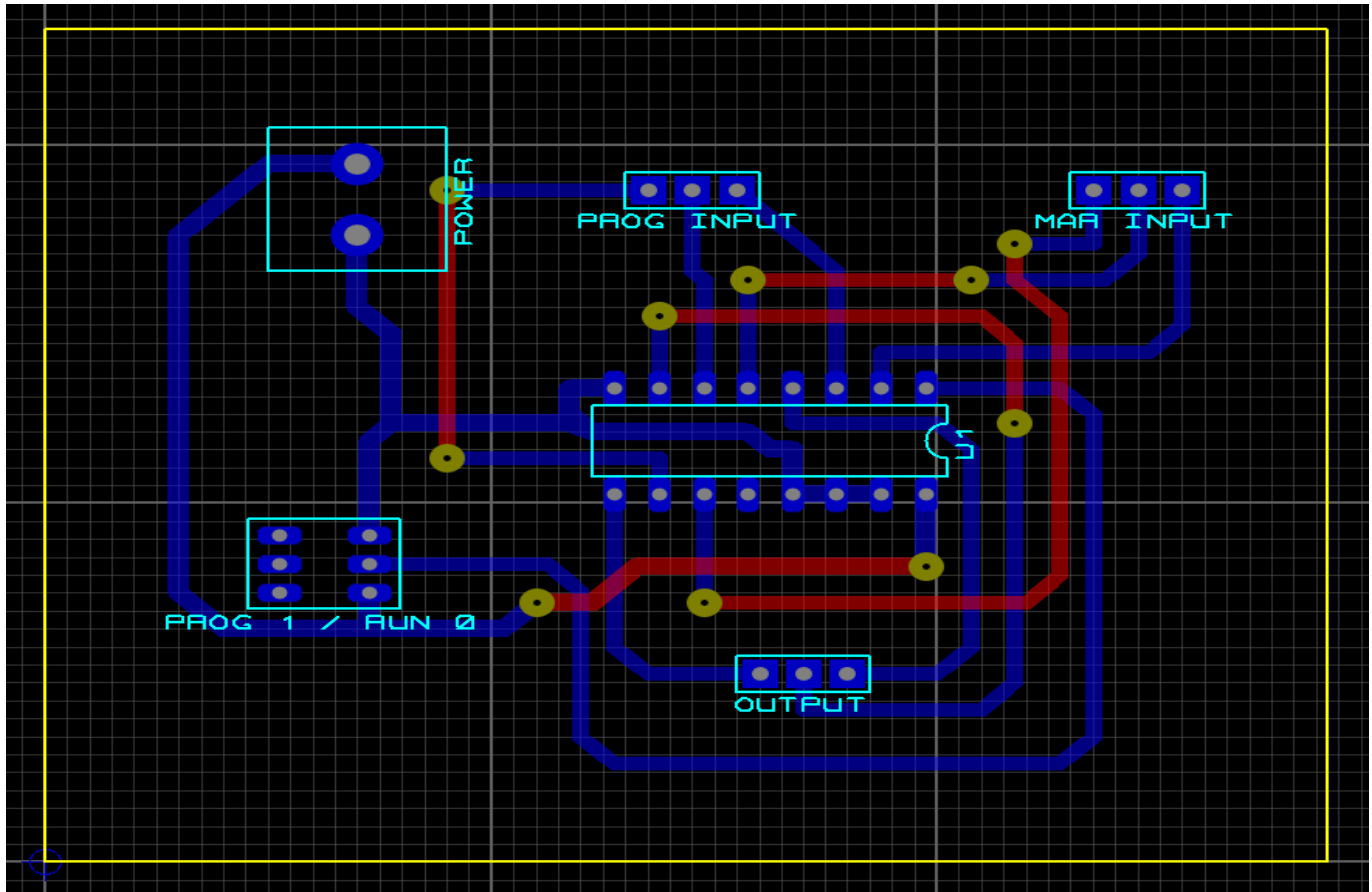


Address Seclector:

Circuit:



PCB:



Problem faced:

- 1) Selecting the proper IC's
- 2) Component placement
- 3) PCB design
- 4) Some components don't have PCB package
- 5) While routing the PCB

Discussion:

In this phase we designed the Registers and memory unit of our SAP. Among all the module RAM designing was the hardest one and output register was the easiest one. It is a 6-bit SAP, we can develop it further to 8 bit, as we same number of components and concepts to built 8 bit SAP. Through this phase we learnt how data is stored and fetched in time need from RAM. We also learnt how memory unit of a computer works. It was a great learning phase for us.

Contribution of Each Member:

ID: 210021330:

RAM was the most complex module among phase A & B. So it took a lot of time to understand its working principle. For designing the RAM I used Two 74HC189, Two 7404 IC (NOT gate) and one 74HC245 IC. As I had to combined Two RAM IC to made a single 8*6 Bit RAM, Pin connection were difficult at first. After completing the circuit for first time it wasn't working, so I had to do some trouble shooting. After trouble shooting the RAM module is working properly.

ID: 210021342:

MAR or memory address register is a register responsible for storing memory address of the data. To complete the construction of MAR, I had to use three control pins like three CONN-SIL-3, one TBLOCK-12 & one 74HC173. At first, I faced some issues regarding the use of 74HC173. Then I went through some youtube videos & found how to do this.

ID: 210021315:

"CLOCK" was one of the most important part among phase A & B. It determines the synchronization of various components within the system. Therefore it has been designed with proper care. For designing the CLOCK I used Two 555 Timer ic, Two 7400 IC (NAND gate) . First I had to design a Astable 555 timer circuit as astable pulse signal then I had to design Monostable 555 timer as selection pin and lastly I had to design Bistable 555 timer as manual pulse signal. Then they were connected by two nand gate as such that the two signals can be controlled by selection pin. As a beginner I faced some problems regarding the use of different switch and pins but as times passes I coped up with that.

ID: 210021318:

The Output Register in the SAP-1 is a part of the Output Unit. It is a special register used to temporarily hold the data that needs to be output to external devices, such as a display or other peripherals. The Output Register is controlled by the control unit of the SAP-1, which manages data transfer between the memory, CPU, and output devices.

When the SAP-1 computer needs to send data to an output device, the data is transferred from the CPU's internal registers or memory into the Output Register. From there, it can be sent to the external device under the control of the control unit.

We used two 74HC173 (4 bit register) and a bunch of resistors to build this circuit. We also used some LEDs to demonstrate the outputs.

ID: 210021312:

The accumulator was not that easy to make at first, after understanding how it is working it started to make sense and then it became doable. I have used two 74173 ICs and a 74HC245 IC. I have also used two 7404 ICs (Not gate). Because of using the ICs the PCB design became quite simpler and easy to make. The circuit is running perfectly and there are no errors showing in both the schematic and PCB part.

ID: 210021305:

The program counter which is a part of the control unit counts from 000 to 111, each combination indicates the binary address the program that is stored in that runs, as for 6-bit sap, I used 74HC181 IC and 74HC245 IC to design program counter. The 3 diode LEDs were used to indicate which memory address it is accessing. The pin diagrams were a bit difficult as I messed up 1st time trying it to do fast but after fixing the connection the program counter worked as it should.

ID: 210021250:

B – Register was the simplest one among all the modules. For making B-Register I used Two 74HC173 IC and one NOT Gate (7404) IC. I faced difficulties in finding proper resources for making this circuit, finally I found one. After fixing some small issues circuit is working correctly.